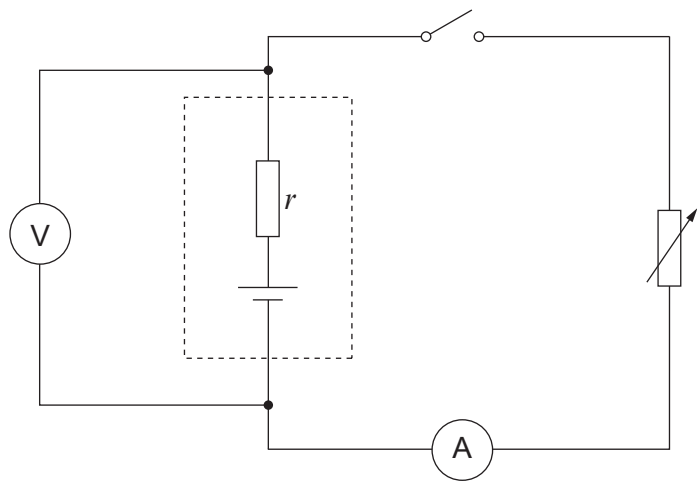
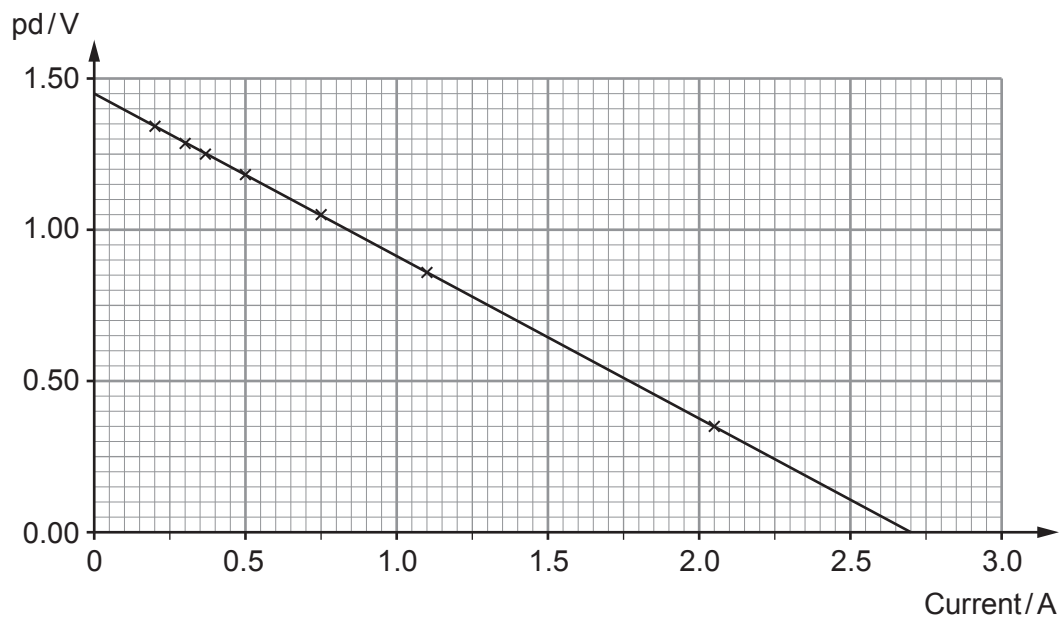


Answer **all** questions.

1. Anwen sets up the circuit shown in order to investigate a cell. The variable resistor is adjusted and readings are obtained from the voltmeter and ammeter.



The readings are plotted as shown.



- (a) The manufacturer claims that the emf of the cell is 1.50V. Explain, in terms of energy, what this statement means. [2]

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(b) An equation applying to a cell of emf, E , and internal resistance, r , is:

$$V = E - Ir$$

(i) Explain how Anwen's graph is in good agreement with this equation. [2]

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(ii) Calculate the gradient of the line and hence determine a value for the cell's internal resistance. [2]

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(iii) Evaluate the manufacturer's claim that the emf is 1.50 V. [1]

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(c) (i) Use the graph to determine the greatest current the cell can supply. [1]

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(ii) State the resistance of the variable resistor when the current is at its maximum. [1]

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